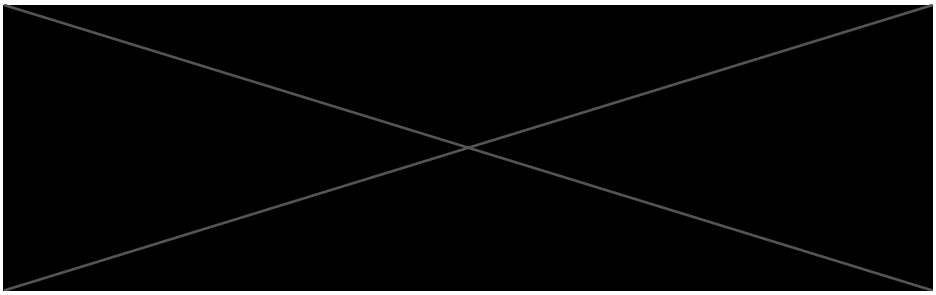
Context Without Contact: Top-Down Information Flow from Shared Modulation in Uncoupled Prebiotic Systems



Abstract

Shifts from bottom-up to top-down influences are considered a hallmark of major evolutionary transitions, including the origin of life. We explore how such directional information flow can arise without direct interactions among the system components using a minimal model of uncoupled logistic maps subject to a shared, time-varying constraint. Transfer entropy (TE) reveals a consistent macro-to-micro bias across diverse modulation profiles, along with increased inter-unit predictability and confinement of macroscopic dynamics to low-dimensional manifolds. These results show that structured environmental variation alone is sufficient to generate an asymmetric top-down directional information flow, suggesting that dynamic external constraints may serve as an early scaffold for coordination before the evolution of internal coupling mechanisms.

Word count: 2,409 words, excluding references.

Keywords: origins of life, emergence, astrobiology, top-down causation

Ethical Compliance: This article does not present research with ethical considerations

Conflict of Interest declaration: The authors declare that they have no affiliations with or involvement in any organization or entity with any financial interest in the subject matter or materials discussed in this Manuscript.

Introduction

Top-down causation, in which higher-level organization shapes the dynamics of lower-level components, occurs at all scales of biological organization and helps explain the hierarchy of structures in living systems (1–3). Indeed, it has been proposed that shifts from predominantly bottom-up to predominantly top-down influences are hallmarks of major evolutionary transitions (4,5), including the emergence of life (6–8). In this view, higher-level organizations emerge from previously independent units and informational control arises at the collective level. One way to study such transitions is by examining the directional information flow between scales, specifically, whether the macroscopic state of a system can predict the dynamics of its parts better than the reverse (9).

This perspective motivates a key question particularly relevant for early chemical evolution: can asymmetric directional information flow between levels arise even in the absence of direct coupling among low-level components, driven solely by a shared environmental structure? Structured external cycles (such as tides, day–night rhythms, or wet–dry fluctuations) are known to promote essential prebiotic processes (10–12), but could such global patterns have also imposed system-wide constraints capable of coordinating dynamics across otherwise independent units?

To probe this, we use a deliberately minimal model in which a population of uncoupled logistic maps is subject to a single time-varying global constraint, implemented as a sinusoidal modulation of the carrying capacity. We then investigated whether such purely environmental structuring could produce measurable macro-to-micro predictive bias. We computed Transfer Entropy (TE) (13) to quantify directional information flow. TE measures how much knowledge of the past of one variable improves predictions about the future of another, beyond what the variable's own past can explain. While the interpretation of informational asymmetry as literal causation remains debated, here we treat it as an operational signal or a measurable statistical signature that may accompany early forms of coordinated behavior, regardless of whether an underlying causal mechanism is in place. This framework has been used to operationalize top-down informational influence in a system where units are explicitly coupled (9); however, here, the

focus is on the limiting case with no internal coupling, isolating environmental structure as a potential driver of coordination in prebiotic contexts.

Results

We consider a minimal system of *N* uncoupled logistic maps, each evolving under a shared time-dependent carrying capacity, K_n. Let $x_{i,n}$ denote the state of unit *i*, where $i = \{1 \ldots N\}$, at discrete time step *n*. Each unit follows:

$$x_{i,n+1} = r_i x_{i,n}\left(1 - \frac{x_{i,n}}{K_n}\right),$$

where $r_i \in [r_{min,} r_{max}]$ is sampled independently from a uniform distribution to ensure that, in the absence of modulation, each unit operates in a chaotic regime (9,14). There are no interactions among the units; the only shared influence is the global carrying capacity K_n, which is modulated sinusoidally as $K_n = K_0 + A\, sin(\omega\, n)$. This modulation represents synchronized environmental rhythms (e.g., day–night cycles, tides, or wet–dry transitions) that affect the entire population. The setup was inspired by an earlier toy model in which top-down control emerged from explicit mean-field coupling (15). In the present case, all units are independent and subject to a common time-varying environment. The macroscopic variable is defined as the ensemble mean: $\mathrm{M}_n = \frac{1}{N}\sum_{i=1}^{N} x_{i,n}$.

While abstract, the model components can be loosely interpreted in prebiotic terms: the global constraint K_n represents an externally imposed limit on the system carrying capacity (e.g., availability of resources, space, or favorable conditions), the intrinsic parameters r_i reflect heterogeneity in how individual entities respond to that environment, and the mean field M_n captures an aggregate property of the population, such as an overall reaction yield or bulk chemical state.

The time-varying K_n imposes a common oscillatory envelope on all trajectories x, introducing a coherent pattern that shapes the ensemble average M (Figure 1A). To assess whether this shared modulation produces directional relationships between macro- and micro-level dynamics, we computed the transfer entropy (TE) in both directions: top-down (from mean field to representative unit, $M \rightarrow x$) and bottom-up (from representative unit to mean field, $x \rightarrow M$). Formally:

$$T_{Y \rightarrow X} = \sum p\left(x_{n+1}, x_n^{(k)}, y_n^{(k)}\right) log_2 \frac{p\left(x_{n+1} | x_n^{(k)}, y_n^{(k)}\right)}{p\left(x_{n+1} | x_n^{(k)}\right)}$$

where $x_n^{(k)}$ and $y_n^{(k)}$ are the histories of length *k* for the target and source variables, respectively.

TE analysis revealed clear directional asymmetry (Figure 1B). Without modulation ($A \approx 0$), TE is negligible in both directions, which is consistent with unstructured dynamics. As *A* increases, the TE from the mean to the individuals ($M \rightarrow x$) increases steadily, whereas the reverse ($x \rightarrow M$) remains negligible. This indicates that shared modulation generates top-down informational influences without reciprocal predictability. The TE was estimated with a histogram-based method using 15 bins and embedding lag k = 2 (rather than the maximum value at k = 1 to reduce near-synchronous correlations that can inflate TE values) (Supplementary Figure S1).

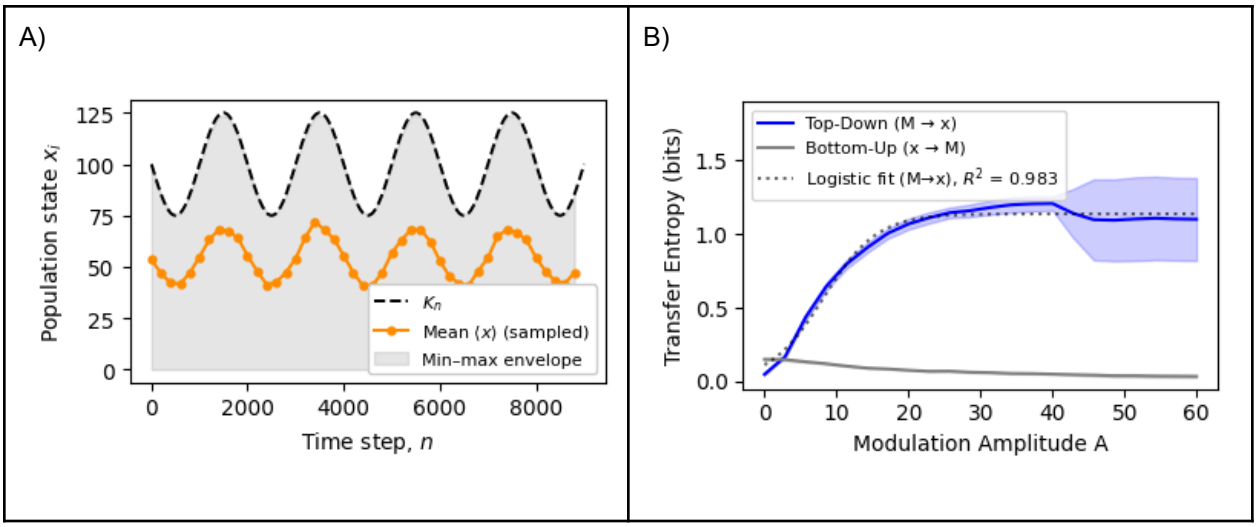


Figure 1. Modulation-driven asymmetry and entrainment under sinusoidal forcing.

(A) Time series of a population of N = 1000 uncoupled logistic units evolving under sinusoidal carrying capacity ($K_0 = 100$, $A = 25$, $r_i \in [3.9, 4.0]$). The black, dashed curve shows the external modulation K_n, the blue line the population mean ⟨x⟩ sampled every 200 steps, and the shaded band the range of individual states at each sample. (B) Transfer entropy (TE) between the population mean M_n and a representative unit $x_{i,n}$, computed in both directions: $M \to x$ (blue) and $x \to M$ (gray), across modulation amplitudes *A*. The dashed line shows the logistic fit to TE($M \to x$). Results are averaged over 10 seeds with *N* = 1,000, $K_0 = 100$, $r_i \in [3.9, 4.0]$, and *T* = 10,000 steps (first 1,000 discarded). Shaded regions show 95% confidence intervals (bootstrap-derived).

At high modulation amplitudes, TE (M→x) approaches a plateau (Figure 1B), well captured by a logistic fit as $TE_{M\to x(A)} = T_{max}/(1 + exp(-\alpha(A - A_0)))$, with R=0.983. This plateau reflects the natural ceiling set by the extent to which a unit's dynamics can be explained by a shared environmental signal. At low amplitudes, idiosyncratic fluctuations from differences in the intrinsic growth rates (r_i) dominate; therefore, the mean field has little predictive power. As the amplitude increases, the common modulation progressively shapes all trajectories, making the mean a stronger predictor of individual behavior. However, once the environmental signal overwhelms individual variability, no further information can be gained, and the predictive relationship has reached the limit imposed by the variance in the environmental

drive itself. This is consistent with the observation that, in the high-amplitude regime, the fraction of variance in individual trajectories explained by the mean field approaches unity (R^2 ~1; Figure 2A). In other words, stronger modulation eventually adds no new structure and rescales only the same signal. The same saturation trend is reflected in the regression-based measures of alignment (Figure 2B), where both the fraction of variance explained and the signal-to-noise ratio (SNR) increase with amplitude before leveling off.

We obtained the same qualitative asymmetry when replacing the sinusoidal forcing with a logistic one (representing gradual favorable environmental shift) and with an inverse logistic (representing gradual environmental decline) (Supplementary Figure S2), indicating that the effect is robust to the shape of the modulation. The saturation was not due to numerical clipping; although a soft bound of (0, $2K_0$) was imposed, it was only approached in the lower end in <10% of the cases at A = 60.

Heuristically, the observed TE asymmetry reflects the growing dominance of shared environmental constraints over independent noise. As A increases, the shared K_n introduces a common pattern into all units, strengthening their correlation with M. Unit-specific fluctuations from intrinsic r values remain uncorrelated and thus average out in the mean. From an information-theoretic standpoint, M is a low-dimensional projection of the global signal, whereas x contains both the signal and idiosyncratic noise. As the signal fraction increases, the predictive information increases in $M \rightarrow x$ whereas the reverse direction remains negligible. The same trend is reflected in the regression-based measures of alignment between the mean field and a representative unit (Figure 2B). Specifically, we quantified alignment using the coefficient of determination (R^2), which represents the fraction of variance in the unit's trajectory explained by the mean, and the signal-to-noise ratio (SNR), defined as the variance of M_n divided by the variance of the regression residuals. Both metrics increase with A, indicating that stronger global modulation makes the mean field a better predictor of individual dynamics.

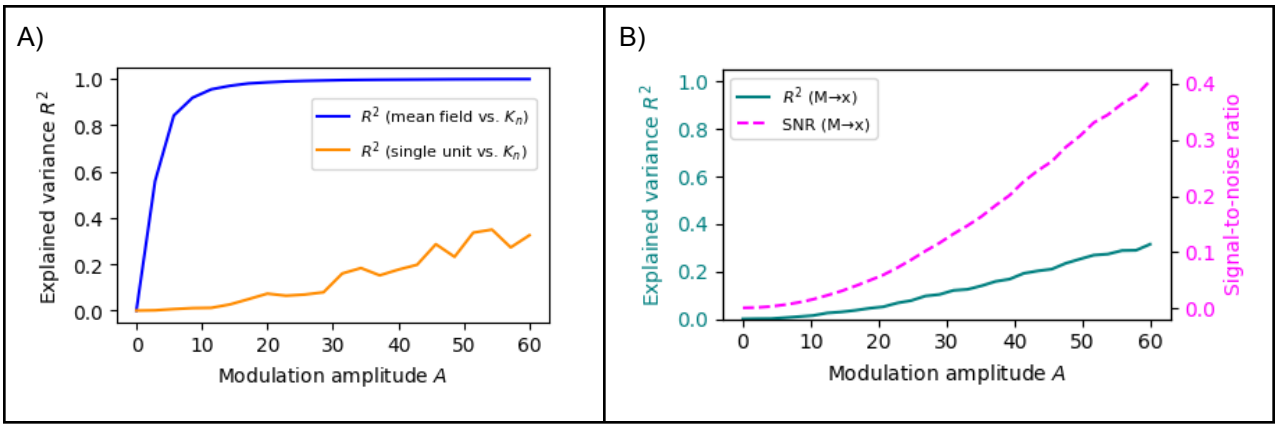


Figure 2. Alignment with mean field and modulation signal under increasing amplitude. (A) Fraction of variance in the mean field (blue) and in a representative unit's trajectory (orange) explained directly by the modulation signal K_n, as a function of A. (B) Relationship between modulation amplitude A and statistical alignment between the population mean field M_n and a representative unit $x_{1,n}$. Alignment is quantified using the coefficient of determination (R^2), which represents the fraction of variance in the unit's trajectory explained by the mean field (teal), and the signal-to-noise ratio (SNR), defined as the variance of M_n divided by the variance of the regression residuals (magenta).

The presence of directional information flow between the mean and individuals raises a further question: can shared modulation generate consistent directional relationships between individuals? Even without interactions, a common drive may induce dependencies across the ensemble. We tested this by comparing the transfer entropy (TE) from a reference unit to 50 other units against a null distribution generated from a circularly shifted surrogate series that preserves autocorrelation while destroying temporal alignment (16) (Figure 3). Significance was defined as a real TE exceeding the 95th percentile of the surrogate distribution, using 100 surrogates per target. With no modulation (A=0), no units exceeded the significance threshold, which is consistent with the absence of temporal coordination. Under moderate modulation (A=25), however, 39 out of 50 units were significant, demonstrating that shared modulation alone induces emergent inter-unit predictability.

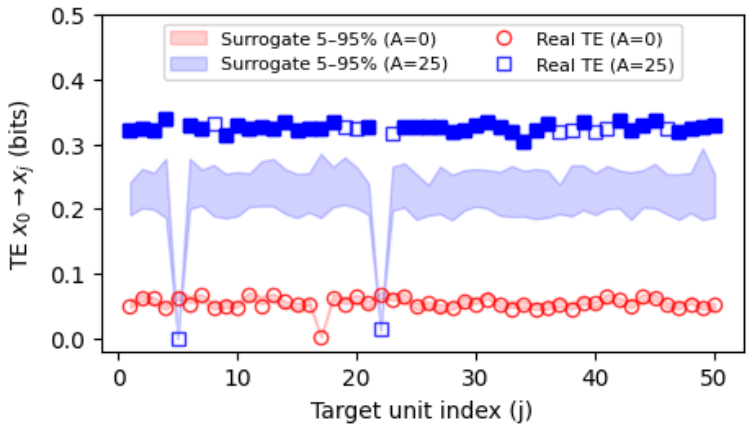


Figure 3. Transfer entropy (TE) from reference unit x_0 to each target unit x_j for two modulation amplitudes ($A = 0$, $A = 25$). Real TE values (symbols) are compared to the distribution from 100 circularly shifted surrogates (shaded bands). Solid symbols mark significant cases ($p < 0.05$); open symbols are non-significant. Results shown for $N = 1000$, $K_0 = 100$, $r_i \in [3.9, 4.0]$, $T = 10,000$ steps (first 1,000 discarded), number of targets analyzed = 50, number of surrogates per target = 100.

Finally, we assessed how modulation shapes macroscopic dynamics by examining the return maps of the mean field. Increasing the modulation amplitude broadens the range of accessible states but maintains trajectories in a coordinated, low-dimensional pattern, indicating that the shared environmental structure expands the dynamical range without disrupting collective organization (Figure S4).

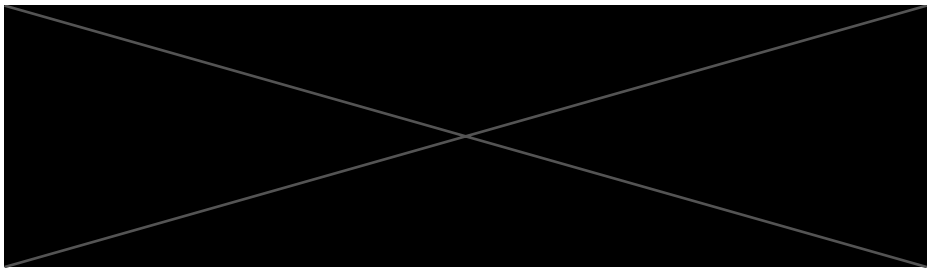
Discussion

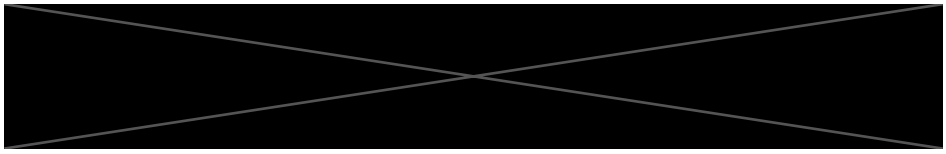
Origins-of-life theories often assume that coordination arises from internal chemical coupling among components (e.g., autocatalytic sets (17), hypercycles (18), and early metabolism (19)). These frameworks share the premise that organizations depend on direct chemical interactions; however, they do not always provide satisfactory explanations for the non-trivial question of how such coupling originates in the first place. Here, we explore a simpler scenario in which structured organizations can emerge without direct interactions, driven solely by shared environmental structures.

Our model isolates this idea in its simplest possible form, as a population of uncoupled logistic maps exposed to the same time-varying constraint. Global modulation, implemented through sinusoidal changes in the carrying capacity, introduces a temporal structure without inducing direct interactions between components. We then used transfer entropy (TE) as an operational measure of top-down informational influence to quantify this asymmetry, a framework previously applied to systems with explicit coupling (9). In this context, TE is not intended to imply mechanistic causation, but to test whether shared environmental modulation alone can induce directional predictability between scales. In other words, we ask whether a 'big picture' view of the system can forecast what the parts will do next, simply because they are all nudged in the same way. Despite the absence of coupling, the system develops directional asymmetry, and the behavior of the macroscopic mean becomes increasingly predictive of individual dynamics, whereas the reverse direction remains negligible. In the transfer entropy (TE) literature, correlations induced by a shared input are often classified as "common driver effects" (20–22) and treated as confounds to be removed (23,24). However, in our context, shared input is the mechanism of interest. Although each unit behaves independently, their independence can be overpowered by a sufficiently strong external driver.

Although the model is minimal and abstract in nature, it directly relates to scenarios in origins-of-life research where prebiotic chemistry is unfolded under rhythmic environmental forcing. Regular cycles such as wet-dry, freeze-thaw, day-night, and tidal rhythms are known to concentrate reactants, drive condensation, promote molecular restructuring, stabilize intermediates, and facilitate protocell assembly (10–12,25–29). Beyond enabling specific reactions, such cycles may impose system-wide constraints, thus aligning otherwise independent chemical processes. We represent these well-studied periodic influences with sinusoidal forcing, however, the results are not limited to perfect periodicity. Logistic and inverse-logistic modulation, which represent gradual environmental improvements or declines, yield comparable qualitative asymmetries. The key factor is not the waveform itself, but its capacity to impose a coherent structure across the system. In our framework, modulated carrying capacity serves as a proxy for this type of global context, acting as a constraint capable of entraining independent components without direct, physical interaction.

This broader principle may apply across many of life's major evolutionary transitions, where collections of simpler units become integrated wholes (in part) due to a shared environment. Structured environmental conditions could have served as an early coordinating force prior to the evolution of internal control mechanisms. For example, vesicle populations may have synchronized division under thermal cycles, early microbial communities may have differentiated along spatial gradients, and hominins may have aligned behaviors through shared cues before developing language. While these examples are speculative, they illustrate how directional information flow can be externally imposed and later internalized by evolving systems. More generally, the findings highlight how shared external drivers can generate proto-organization in uncoupled systems, a mechanism that may recur across the physical, chemical, and biological domains. Our results suggest that such environmentally driven coordination, even if transient, may provide a scaffold for the emergence of persistent self-organized complexity.





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